Rank Transform of an Array

Problem Statement

Given an array `arr` containing integers, replace each element with its `rank` in the array.

- The `rank` is determined based on the `size of the element` compared to others in the array.
- Smaller numbers get a lower rank, and equal numbers share the same rank.
- The ranking starts from 1.

Examples

Example 1:

- **Input**: `[10, 20, 20, 30]`
- **Expected Output**: `[1, 2, 2, 3]`
- **Justification**: 10 is the smallest, so its rank is 1. 20, occurring twice, shares the rank 2. 30, being the largest, gets the rank 3.

Example 2:

- **Input**: `[100, 2, 70, 2]`
- **Expected Output**: `[3, 1, 2, 1]`
- **Justification**: 2, being the smallest, is ranked 1. 70 is next, so it's ranked 2. 100, the largest, is ranked 3.

Example 3:

- **Input**: `[5, 5, 5, 5]`
- **Expected Output**: `[1, 1, 1, 1]`
- **Justification**: All elements are the same, so they all share the same rank, 1.

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

public class Solution {
    // Function to transform the array into its rank form
    public int[] arrayRankTransform(int[] arr) {
        // ToDo: Write Your Code Here.
        return arr;
    }
}
```

Test Case | Run Code Result | Stdout

[40,10,20,30]

Use Example Testcases | Run | Submit